

Mahdi Islam

 mahdiislam79.github.io

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EDUCATION

Erasmus Mundus Joint Master: Medical Imaging & Applications (MAIA) September 2023 – September 2025
University of Girona (Spain) | University of Cassino (Italy) | University of Bourgogne (France) **Grade: 8.96/10.00**

Relevant Courses: Advanced Image Analysis, Machine & Deep Learning, Medical Image Segmentation & Applications, Medical Image Registration & Applications, Computer Aided Diagnosis

Thesis: Uncertainty-Guided Active Learning for Access Route Segmentation and Planning in Transcatheter Aortic Valve Implantation

Bachelor of Science: Electrical & Electronic Engineering

January 2018 – May 2022

Islamic University of Technology (IUT), Bangladesh

Grade: 3.63/4.00

RESEARCH & WORK EXPERIENCE

Lecturer, Department of Electrical & Electronic Engineering

Independent University, Bangladesh (IUB)

January 2026 – Present

- Teaching undergraduate courses in Electrical & Electronic Engineering; supervising student research projects and mentoring thesis students toward research publications.

Research Supervisor, Biomedical Instrumentation and Signal Processing Lab Wing

Center for Computational & Data Sciences (CCDS)

Independent University, Bangladesh (IUB)

January 2026 – Present

Research Assistant, Department of Radiology

Medical University of Innsbruck, Austria

February 2025 – July 2025

- Implemented active learning strategies for an annotation-efficient 3D segmentation pipeline, followed by vessel diameter measurement, to extract iliac diameter profiles for access-route planning in Transcatheter Aortic Valve Implantation (TAVI).
- Developed a multimodal pipeline integrating deep learning and imageomics from pre-TAVI CT and CMR scans to predict post-procedural mortality, with interpretability supported by SHAP analysis and survival insights from Kaplan-Meier analysis.

Lecturer, Department of Computer Science & Engineering

Metropolitan University, Bangladesh

August 2022 – June 2023

- Taught core CS/EEE courses; introduced machine learning concepts and mentored sophomore students, contributing to increased student engagement with thesis-based research.

Undergraduate Research Assistant, Department of Electrical & Electronic Engineering

Islamic University of Technology (IUT), Bangladesh

April 2021 – April 2022

- Created a novel dataset from sensor-acquired foot sole heat-map videos using image augmentation techniques.
- Developed a real-time multiclass classification model to detect types of gait anomalies using Keras and TensorFlow.

PUBLICATIONS

- M. Islam**, M. Tabassum. “Brain Metastases Segmentation for BraTS 2026 Task 1: A Multi-Architecture Comparison” **BraTS-METS, MICCAI 2026** [Code] | Conference Paper
Keywords: Brain Metastases · Tumor Segmentation · Transformer · Data Leakage · Small Lesion Detection
- E. Almar-Muñoz, **M. Islam**, M. Tabassum, M. Haltmeier, M. Pamminger, L. Neumann, C. Kremser, M. Reindl, S. J. Reinstadler, B. Metzler, A. Mayr. “Automated Aortoiliac femoral Segmentation and Transfemoral Access Planning Using Non-Contrast MR Angiography for Transcatheter Aortic Valve Replacement Guidance.” **Radiology: Cardiothoracic Imaging**, 2026 [Under Review] | Journal Article
Keywords: MR Angiography · Aortoiliac femoral Segmentation · Transfemoral Access Planning · TAVR Guidance · Deep Learning
- J. Bai, Y. Tang, Z. Zhou, **M. Islam**, M. Tabassum, E. Almar-Muñoz, *et al.* “FUGC: Benchmarking Semi-Supervised Learning Methods for Cervical Segmentation.” **IEEE Transactions on Medical Imaging**, 2026 [DOI] | Journal Article
Keywords: Cervical Segmentation · Semi-Supervised Learning · Vision Transformer · Fetal Ultrasound
- M. Islam**, M. Tabassum, A. Mayr, C. Kremser, M. Haltmeier, E. Almar-Muñoz. “Uncertainty-Guided Active Learning for Access Route Segmentation and Planning in Transcatheter Aortic Valve Implantation.” **Journal of Imaging, (MDPI)**, 2025 [DOI] | Journal Article
Keywords: Active Learning · Vessel Diameter Quantification · Aortic Segmentation · TAVI Planning · CMR

- [5] **M. Islam**, M. Tabassum, M. Elbatel, A. Mayr, C. Kremser, M. Haltmeier, E. Almar-Muñoz. “*Semi-Supervised Transformer-Based Cervical Segmentation: FUGC 2025 Challenge.*” **IEEE 22nd International Symposium on Biomedical Imaging (ISBI)**, 2025 [DOI] | Conference Paper
Keywords: Fetal Ultrasound · Semantic Segmentation · Weak-to-Strong Consistency · Vision Transformer
- [6] **M. Islam**, M. Tabassum, M. M. Nishat, F. Faisal, M. S. Hasan. “*Real-Time Clinical Gait Analysis and Foot Anomalies Detection Using Pressure Sensors and Convolutional Neural Network.*” **International Conference on Business and Industrial Research (ICBIR)**, 2022 [DOI] | Conference Paper
Keywords: Pressure Sensors · Transfer Learning · Gait Analysis · Foot Anomalies · CNN

PROJECTS

Multimodal Outcome Prediction for Post-TAVI Mortality Using Cardiac CT and CMR | [Link](#)

Medical University of Innsbruck

March 2025 – July 2026

- Built an outcome-prediction pipeline using cardiac CT and CMR data to estimate post-TAVI mortality risk.
- Extracted segmentation-based imageomics features; applied SHAP and correlation analysis to identify anatomical predictors of adverse outcomes.
- Explored self-supervised pretraining (VoCo, masked autoencoders) for domain-specific representations under limited labeled data.

Curvature-Aware Parameter-Efficient Fine-Tuning for Vision–Language Models | [Link](#)

Remote Research Collaboration

July 2025 – Present

- Extended LoRA with Fisher-information-based (K-FAC-style) curvature approximations to stabilize low-rank adaptation in large VLMs.
- Implemented adaptive rank reprojection and rank-stabilized LoRA to dynamically control model capacity under limited supervision.

Deep Learning-Based Brain Tissue Segmentation Using U-Net and MRI Data | [Link](#)

University of Girona

November 2024 – January 2025

- Developed a U-Net model for CSF, GM, and WM segmentation from [IBSR18](#) MRI scans, with advanced preprocessing, augmentation, and one-hot encoding.
- Achieved strong Dice scores using a PyTorch pipeline with Weights & Biases for experiment tracking and hyperparameter optimization.

Atlas and Tissue-Model Guided Gaussian Mixture Models for Brain Tissue Segmentation | [Link](#)

University of Girona

September 2024 – November 2024

- Developed a hybrid segmentation approach combining Gaussian Mixture Models (GMM) with probabilistic atlas-based methods, integrating spatial and intensity information for gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF) segmentation from MRI scans.
- Implemented multiple initialization strategies (K-means, Tissue Model) within the Expectation-Maximization (EM) algorithm, achieving improved Dice scores over traditional methods.

Skin Cancer Detection from Dermoscopic Images: Comparing ML & DL Approaches | [Link](#)

University of Girona

September 2024 – January 2025

- Built end-to-end ML and DL pipelines for binary and multiclass skin lesion classification from dermoscopic images, including preprocessing (hair removal, color normalization, ROI extraction) and feature extraction using GLCM, LBP, Gabor filters, and HOG.
- Implemented SVM, Random Forest, and XGBoost classifiers alongside EfficientNet-based deep learning models, addressing class imbalance through resampling techniques.

4D Chest CT Volume Registration: DIR-Lab Challenge | [Link](#)

University of Girona

November 2024 – January 2025

- Developed a baseline deformable registration model using elastix for thoracic CT alignment across respiratory phases in COPD patients.
- Built a deep learning pipeline using VoxelMorph to enhance registration accuracy and benchmark against the baseline.

Colorectal Cancer Tissue Classification and Gland Segmentation from Histopathology Images | [Link](#)

University of Cassino

March 2024 – May 2024

- Built a classical segmentation pipeline (K-means, Watershed) and a ML classification pipeline using GLCM, LBP, and Gabor features with XGBoost, LightGBM, and SVM.
- Engineered a deep learning segmentation pipeline in PyTorch with UNet/UNet++ and VGG16, ResNet, EfficientNet backbones.

ACCOMPLISHMENTS

Fetal Ultrasound Grand Challenge (FUGC) | ISBI 2025 | [Certificate](#) March 2025

- Ranked **5th** **internationally** among all competing teams in this IEEE-organized medical image segmentation challenge.

Intherapy Graduate School Scholarship | University of Bourgogne, France | [Certificate](#) January 2025
Associate Data Scientist Certificate | DataCamp | [Certificate](#) November 2024

EXTRACURRICULAR ACTIVITIES

Coordinator – IT & Web Design, Erasmus Mundus Association Bangladesh January 2026 – Present
Deputy Coordinator – IT & Web Design, Erasmus Mundus Association Bangladesh January 2025 – December 2025
Captain, IUT University Tennis Club April 2021 – May 2022

SKILLS

Programming: Python, R, MATLAB, C++, SQL, JavaScript | **DL Architectures:** CNNs, Transformers, LLMs, VLMs, LSTMs, RNNs
Frameworks: PyTorch, MONAI, TensorFlow, Hugging Face, Scikit-learn | **Medical Tools:** ITK-SNAP, 3D Slicer, FreeSurfer, ITK-Elastix, SPM
Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Scikit-image | **Compute:** Distributed Training (DDP), Multi-GPU HPC Clusters

RESEARCH INTERESTS

Explainable & Trustworthy AI for Medical Imaging, Self-Supervised Learning & Large-Scale Pretraining, Uncertainty Estimation, Multimodal Data Fusion, Medical Image Segmentation & Analysis, Parameter-Efficient Fine-Tuning of Foundation Models, Radiomics & Biomarker Analysis for Early Disease Detection, Prognosis, and Individualized Risk Prediction.

STANDARDIZED TEST SCORES

IELTS: Listening 9 | Reading 9 | Speaking 7 | Writing 7 | **Overall 8**

REFERENCES

Dr. Alessandro Bria, Associate Professor | Course Instructor and Project Supervisor
Department of Electrical and Information Engineering
University of Cassino and Southern Latium, 03043 Cassino, Italy
Email: a.bria@unicas.it

Fahim Faisal, Researcher | Bachelor's Thesis Supervisor
Department of Neuromedicine and Movement Science
NTNU, 7491 Trondheim, Norway
On leave from: Assistant Professor, Department of Electrical and Electronic Engineering
Islamic University of Technology, Gazipur, Bangladesh
Email: fahim.faisal@ntnu.no